

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
root@DESKTOP-UU3FNJ4:~# nano prog001.sh
root@DESKTOP-UU3FNJ4:~# chmod +x prog001.sh
root@DESKTOP-UU3FNJ4:~# ./prog001.sh
Parent PID: 655
Child PID: 656
  PID  PPID  CMD
  636   635  -bash
  655   636  /bin/bash ./prog001.sh
  656   655  sleep 20
  657   655  ps -o pid,ppid,cmd
Child processes of this shell:
  PID  TTY      TIME  CMD
  656  pts/2    00:00:00 sleep
  658  pts/2    00:00:00 ps
root@DESKTOP-UU3FNJ4:~# nano prog001.sh
root@DESKTOP-UU3FNJ4:~# sleep 30
root@DESKTOP-UU3FNJ4:~# sleep 30
^Z
[1]+  Stopped                  sleep 30
root@DESKTOP-UU3FNJ4:~# sleep 50
^Z
[2]+  Stopped                  sleep 50
root@DESKTOP-UU3FNJ4:~# pstree -p $$
bash(636)─pstree(681)
          │sleep(675)
          └─sleep(678)
root@DESKTOP-UU3FNJ4:~# pstree -p
systemd(1)─agetty(200)
           │agetty(222)
           └─cron(187)
              └─dbus-daemon(188)
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
root@DESKTOP-UU3FNJ4:~# sleep 300
^Z
[4]+  Stopped                  sleep 300
root@DESKTOP-UU3FNJ4:~# cat
^Z
[5]+  Stopped                  cat
root@DESKTOP-UU3FNJ4:~# jobs
[1]  Stopped                  sleep 30
[2]  Stopped                  sleep 50
[3]  Stopped                  sleep 500
[4]- Stopped                  sleep 300
[5]+ Stopped                  cat
root@DESKTOP-UU3FNJ4:~# bg %2
[2] sleep 50 &
root@DESKTOP-UU3FNJ4:~# jobs
[1]  Stopped                  sleep 30
[2]  Done                    sleep 50
[3]  Stopped                  sleep 500
[4]- Stopped                  sleep 300
[5]+ Stopped                  cat
root@DESKTOP-UU3FNJ4:~# fg %3
sleep 500
cat
i am running in foreground
i am running in foreground
^Z
[3]+  Stopped                  sleep 500
root@DESKTOP-UU3FNJ4:~# jobs
[1]  Stopped                  sleep 30
[3]+ Stopped                  sleep 500
[4]  Stopped                  sleep 300
```



```
root@DESKTOP-UU3FNJ4: ~  
GNU nano 7.2 prog001.sh  
#!/bin/bash  
  
echo "Parent PID: $$"  
  
sleep 20 &  
echo "Child PID: $!"  
  
ps -o pid,ppid,cmd  
  
echo "Child processes of this shell:"  
ps --ppid $$  
  
[ Read 11 lines ]  
^G Help      ^O Write Out  ^W Where To   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Repl:     ^U Paste      ^J Justify  
^C Location   M-U Undo      M-A Set Mark  
^_/ Go To Line M-E Redo      M-6 Copy
```

```
root@DESKTOP-UU3FNJ4: ~  
root@DESKTOP-UU3FNJ4:~# nano prog002.sh  
root@DESKTOP-UU3FNJ4:~# chmod +x prog002.sh  
root@DESKTOP-UU3FNJ4:~# ./prog002.sh  
  
./prog002.sh: line 1: quot: command not found  
./prog002.sh: line 1: job: command not found  
./prog002.sh: line 1: quot: command not found  
Wed Apr  8 17:55:11 UTC 2026  
  
./prog002.sh: line 5: quot: command not found  
./prog002.sh: line 5: job: command not found  
./prog002.sh: line 5: quot: command not found  
root  
  
./prog002.sh: line 9: quot: command not found  
./prog002.sh: line 9: job: command not found  
./prog002.sh: line 9: quot: command not found  
./prog002.sh: line 10: cal: command not found  
  
./prog002.sh: line 13: quot: command not found  
./prog002.sh: line 13: job: command not found  
./prog002.sh: line 13: quot: command not found  
  
a.txt          aksharidsa    archive.tar    f1.sh          f7.sh          file2.txt.gz   prog1.sh  
aks            aksharigowda bca            f1.txt         f9.sh          fruit.txt      prog2.sh  
aksh           akshariu      bca.txt       f10.sh         file            fruit1         ps1.sh  
'aksh gowda'   akshariush    bca1          f11.sh         file.txt        hello          stu_marks.tx  
aksha          akshariushankar bca2          f12.sh         file.txt.gz     merged.txt     stu_name.tx  
akshari        archieve.cpio  bca3          f13.sh         file1.gz        notes.txt      student.txt  
'akshari gowda' archieve.tar   combined.txt   f3.sh          file12          prog001.sh  
'akshari u shankar' archieve.tar   dir2          f4.sh          file2           prog002.sh  
'akshari u shankar cb ' archieve.cpio  f1            f5            file2.txt      prog01.sh
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
'akshari u shankar cb ' archive.cpio f1 f5 file2.txt prog01.sh

./prog002.sh: line 17: quot: command not found
./prog002.sh: line 17: quot: command not found
./prog002.sh: line 17: job: command not found

./prog002.sh: line 18: quot: command not found
./prog002.sh: line 18: bg: no job control
./prog002.sh: line 18: quot: command not found

./prog002.sh: line 21: quot: command not found
./prog002.sh: line 21: quot: command not found
/root

./prog002.sh: line 21: job: command not found
./prog002.sh: line 24: quot: command not found
./prog002.sh: line 24: All: command not found
./prog002.sh: line 24: quot: command not found
root@DESKTOP-UU3FNJ4:~# ./lab3.sh
-bash: ./lab3.sh: No such file or directory
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
GNU nano 7.2 prog002.sh
echo &quot;job 1: Display Date&quot;;
date
sleep 3

echo &quot;job 2: Display current user&quot;;
whoami
sleep 3

echo &quot;job 3: Display Calender&quot;;
cal
sleep 2

echo &quot; job 4: list the files&quot;;
ls
sleep 3

echo &quot;job 5: display my system date&quot;;
date +%&quot;%T&quot;;
sleep 1

echo &quot;job 6: print working directory&quot;;
pwd

echo &quot;All jobs are finished&quot;;

[ Read 24 lines ]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-G Copy
```

```
root@DESKTOP-UU3FNJ4: ~  
-bash: ./lab3.sh: No such file or directory  
root@DESKTOP-UU3FNJ4:~# nano prog003.sh  
root@DESKTOP-UU3FNJ4:~# chmod +x prog003.sh  
root@DESKTOP-UU3FNJ4:~# ./prog003.sh  
Job 1: Display Date  
Wed Apr 8 17:57:18 UTC 2026  
Job 2: Display current user  
root  
Job 3: Display Calendar  
./prog003.sh: line 12: cal: command not found  
Job 4: List the files  
a.txt          aksharidsa      archive.tar      f1.sh          f7.sh          file2.txt.gz    prog01.sh  
aks            aksharigowda    bca              f1.txt         f9.sh          fruit.txt       prog1.sh  
aksh           akshari          bca.txt         f10.sh        file           fruit1         prog2.sh  
'aksh gowda'   aksharius       bca1            f11.sh        file.txt       hello          ps1.sh  
aksha          akshariush      bca2            f12.sh        file.txt.gz    merged.txt     stu_marks.tx  
akshari        akshariushankar bca3            f13.sh        file1.gz       notes.txt      stu_name.tx  
'akshari gowda' archieve.cpio    combined.txt     f3.sh         file12         prog001.sh     student.txt  
'akshari u shankar' archieve.tar     dir2            f4.sh         file2          prog002.sh  
'akshari u shankar cb ' archive.cpio      f1              f5             file2.txt      prog003.sh  
Job 5: Display system time  
17:57:29  
Job 6: Print working directory  
/root  
All jobs are finished  
root@DESKTOP-UU3FNJ4:~# sudo apt update  
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease  
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1578 kB]
```

```
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1876 kB]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [252 kB]  
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]  
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [342 kB]  
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]  
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1167 kB]  
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]  
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]  
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1663 kB]  
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]  
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]  
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [322 kB]  
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.8 kB]  
Get:19 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2763 kB]  
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]  
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.2 kB]  
Get:22 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [644 kB]  
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2900 kB]  
Get:24 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]  
Get:25 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [544 B]  
Get:26 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6980 B]  
Get:27 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]  
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [673 kB]  
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]  
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]  
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]  
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7520 B]  
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]  
Get:34 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [500 B]  
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]  
Get:36 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7380 B]
```

```
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:34 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [500 B]
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:36 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7380 B]
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:41 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1480 B]
Get:42 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:43 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [780 B]
Get:44 http://archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [372 B]
Get:45 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 15.7 MB in 1min 14s (213 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
180 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@DESKTOP-UU3FNJ4:~# sudo apt install bsdmainutils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncal
Suggested packages:
  calendar whois vacation mailutils
The following NEW packages will be installed:
  bsdmainutils ncal
0 upgraded, 2 newly installed, 0 to remove and 180 not upgraded.
Need to get 24.2 kB of archives.
After this operation, 76.8 kB of additional disk space will be used.
```

```
root@DESKTOP-UU3FNJ4: ~
After this operation, 76.8 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 ncal amd64 12.1.8 [21.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 bsdmainutils all 12.1.8 [3176 B]
Fetched 24.2 kB in 3s (7986 B/s)
Selecting previously unselected package ncal.
(Reading database ... 40792 files and directories currently installed.)
Preparing to unpack .../archives/ncal_12.1.8_amd64.deb ...
Unpacking ncal (12.1.8) ...
Selecting previously unselected package bsdmainutils.
Preparing to unpack .../bsdmainutils_12.1.8_all.deb ...
Unpacking bsdmainutils (12.1.8) ...
Setting up ncal (12.1.8) ...
Setting up bsdmainutils (12.1.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
root@DESKTOP-UU3FNJ4:~# ./prog003.sh
Job 1: Display Date
Wed Apr  8 18:00:08 UTC 2026
Job 2: Display current user
root
Job 3: Display Calendar
April 2026
Su Mo Tu We Th Fr Sa
      1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30
Job 4: List the files
a.txt          aksharidsa      archive.tar      f1.sh          f7.sh          file2.txt.gz    prog01.sh
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
26 27 28 29 30
Job 4: List the files
a.txt          aksharidsa    archive.tar   f1.sh         f7.sh         file2.txt.gz  prog01.sh
aks            aksharigowda bca           f1.txt        f9.sh         fruit.txt     prog1.sh
aksh          akshariu     bca.txt       f10.sh        file          fruit1        prog2.sh
'aksh gowda'  aksharius    bca1          f11.sh        file.txt      hello         ps1.sh
aksha         akshariush   bca2          f12.sh        file.txt.gz   merged.txt    stu_marks.tx
akshari       akshariushankar bca3         f13.sh        file1.gz      notes.txt     stu_name.tx
'akshari gowda'  achieve.cpio combined.txt  f3.sh         file12        prog001.sh    student.txt
'akshari u shankar'  archive.tar  dir2          f4.sh         file2         prog002.sh
'akshari u shankar cb ' archive.cpio  f1            f5            file2.txt     prog003.sh
Job 5: Display system time
18:00:19
Job 6: Print working directory
/root
All jobs are finished
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
GNU nano 7.2 prog003.sh
#!/bin/bash

echo "Job 1: Display Date"
date
sleep 3

echo "Job 2: Display current user"
whoami
sleep 3

echo "Job 3: Display Calendar"
cal
sleep 2

echo "Job 4: List the files"
ls
sleep 3

echo "Job 5: Display system time"
date +"%T"
sleep 1

echo "Job 6: Print working directory"
pwd

echo "All jobs are finished"

[ Read 26 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
^C Location   M-U Undo     M-A Set Mark
^_/ Go To Line M-E Redo     M-6 Copy
```

```
root@DESKTOP-UU3FNJ4: ~  
root@DESKTOP-UU3FNJ4:~# chmod +x prog004.sh  
root@DESKTOP-UU3FNJ4:~# ./prog004.sh  
----- FILE OPERATIONS -----  
1. Create File  
2. Copy File  
3. Move File  
4. Delete File  
5. Display File  
6. Exit  
Enter your choice:1  
Enter filename to ceate:cbakshari  
Enter content:i am newly created file  
File 'cbakshari' created successfully.  
  
----- FILE OPERATIONS -----  
1. Create File  
2. Copy File  
3. Move File  
4. Delete File  
5. Display File  
6. Exit  
Enter your choice:5  
Enter filename to display:cbakshari  
Content of 'cbakshari':  
i am newly created file  
  
----- FILE OPERATIONS -----  
1. Create File  
2. Copy File  
3. Move File  
4. Delete File
```

```
5. Display File  
6. Exit  
Enter your choice:2  
Enter source filename:cbakshari  
Enter destination filename:dev  
File copied successfully.  
  
----- FILE OPERATIONS -----  
1. Create File  
2. Copy File  
3. Move File  
4. Delete File  
5. Display File  
6. Exit  
Enter your choice:5  
Enter filename to display:dev  
Content of 'dev':  
i am newly created file  
  
----- FILE OPERATIONS -----  
1. Create File  
2. Copy File  
3. Move File  
4. Delete File  
5. Display File  
6. Exit  
Enter your choice:3  
Enter filename to move:dev  
Enter new filename:cbakshari  
mv: cannot stat '&dev': No such file or directory  
File moved successfully.
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
File moved successfully.

----- FILE OPERATIONS -----
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice:4
Enter filename to delete:cbakshari
File deleted successfully.

----- FILE OPERATIONS -----
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice:5
Enter filename to display:cbakshari
Content of 'cbakshari':
cat: cbakshari: No such file or directory

----- FILE OPERATIONS -----
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File

6. Exit
Enter your choice:6
Exiting program...
root@DESKTOP-UU3FNJ4:~# nano prog005.sh
root@DESKTOP-UU3FNJ4:~# chmod +x prog005.sh
root@DESKTOP-UU3FNJ4:~# ./prog005.sh
---- File Permission & Ownership ----
Enter file name:
cbakshari
File not found!
root@DESKTOP-UU3FNJ4:~# nano prog005.sh
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
GNU nano 7.2 prog004.sh
#!/bin/bash
while true
do
echo ----- FILE OPERATIONS -----
echo 1." Create File"
echo 2. "Copy File"
echo 3. "Move File"
echo 4. "Delete File"
echo 5." Display File"
echo 6. "Exit"
echo -n "Enter your choice:"
read choice
if [ $choice -eq 1 ]; then
read -p "Enter filename to ceate:" fname # fname - variable
if [ -f $fname ]; then # $fname - value stored inside the variable
echo "Error: File already exists!"
else
read -p "Enter content:" content
echo "$content">"$fname"
echo "File '$fname' created successfully."
fi
elif [ $choice -eq 2 ]; then
read -p "Enter source filename:" source
read -p "Enter destination filename": dest
if [ -f "$source" ]; then
cp "$source" "$dest"
[ Read 65 lines ]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-6 Copy
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
GNU nano 7.2 prog004.sh
if [ -f "$source" ]; then
cp "$source" "$dest"
echo "File copied successfully."
else
echo "Error: Source file does not exist!"
fi
elif [ $choice -eq 3 ]; then
read -p "Enter filename to move:" source
read -p "Enter new filename:" dest
if [ -f "$source" ]; then
mv "$source" "$dest"
echo "File moved successfully."
else
echo "Error: File does not exist!"
fi
elif [ $choice -eq 4 ]; then
read -p "Enter filename to delete:" fname
if [ -f $fname ]; then
rm "$fname"
echo "File deleted successfully."
else
echo "Error: File does not exist!"
fi
elif [ $choice -eq 5 ]; then
read -p "Enter filename to display:" fname
if [ -f "$fname" ]; then
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-6 Copy
```

```

echo "Content of '$fname':"
cat "$fname"
else
echo "Error: File does not exist!"
fi
elif [ $choice -eq 6 ]; then
echo "Exiting program..."
break
else
echo "Invalid choice!"
fi
echo ""
done

```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^/_ Go To Line	M-E Redo	M-6 Copy

```

root@DESKTOP-UU3FNJ4: ~
root@DESKTOP-UU3FNJ4:~# nano prog005.sh
root@DESKTOP-UU3FNJ4:~# chmod +x prog005.sh
root@DESKTOP-UU3FNJ4:~# ./prog005.sh
---- File Permission & Ownership ----
Enter file name:
dev
Enter permission (e.g., 755):
755
Enter new owner name:
adhya
chown: invalid user: 'adhya'
Updated the details:
-rwxr-xr-x 1 root root 24 Apr 10 05:45 dev

```

```

GNU nano 7.2                                prog005.sh
#!/bin/bash

```

```

echo "---- File Permission & Ownership ----"

# Get filename
echo "Enter file name:"
read file

# Check file exists
if [ ! -f "$file" ]; then
echo "File not found!"

exit
fi

# Get permission
echo "Enter permission (e.g., 755):"
read perm

# Change permission
chmod $perm $file

# Get owner
echo "Enter new owner name:"
read owner
#change ownership
chown $owner $file

```

[Read 30 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^/_ Go To Line	M-E Redo	M-6 Copy

```
#Display new permissions
echo "Updated the details:"
ls -l $file
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-6 Copy

```
root@DESKTOP-UU3FNJ4: ~  
root@DESKTOP-UU3FNJ4:~# nano prog006.sh  
root@DESKTOP-UU3FNJ4:~# chmod +x prog006.sh  
root@DESKTOP-UU3FNJ4:~# ./prog006.sh  
Lab 6:Directory Management
```

1. Create Directory
2. List Directory
3. Delete Directory
4. Exit

Enter your choice:

1

Enter directory name to create:

aksharicb

./prog006.sh: line 17: [n: command not found

Error creating directory.

1. Create Directory
2. List Directory
3. Delete Directory
4. Exit

Enter your choice:

2

Directories in current location:

aks/	aksharidsa/	bca2/
'aksh gowda/'	aksharigowda/	bca3/
aksh/	akshariu/	dir2/
aksha/	aksharius/	f1.txt/
akshari gowda/'	akshariush/	f5/
akshari u shankar cb /'	akshariushankar/	file/
'akshari u shankar/'	bca/	file12/
akshari/	bca1/	hello/

```
root@DESKTOP-UU3FNJ4: ~  
root@DESKTOP-UU3FNJ4: ~  
akshari/                      bca1/                      hello/
```

1. Create Directory
2. List Directory
3. Delete Directory
4. Exit

Enter your choice:

3

Enter directory name to delete:

aks

rmdir: failed to remove 'aks': Directory not empty

Error deleting directory (may not be empty).

1. Create Directory
2. List Directory
3. Delete Directory
4. Exit

Enter your choice:

4

Exiting program...

```
GNU nano 7.2                                prog006.sh
#!/bin/bash
echo "Lab 6:Directory Management"
while true
do
    echo ""
    echo "1. Create Directory"
    echo "2. List Directory"
    echo "3. Delete Directory"
    echo "4. Exit"
    echo "Enter your choice:"
    read choice
    case $choice in
1)
        echo "Enter directory name to create:"
        read dirname
        if [n $? -eq 0 ]; then
            echo "Directory '$dirname' created successfully."
        else
            echo "Error creating directory."
        fi
        ;;
2)
        echo "Directories in current location:"
        ls -d */
        ;;
3)
        echo "Enter directory name to delete:"
        read dirname
        rmdir "$dirname"
        if [ $? -eq 0 ]; then
            echo "Directory '$dirname' deleted successfully."
        else
            echo "Error deleting directory (may not be empty)."

|         |              |             |          |            |               |          |              |
|---------|--------------|-------------|----------|------------|---------------|----------|--------------|
| ^G Help | ^O Write Out | ^W Where Is | ^K Cut   | ^T Execute | ^C Location   | M-U Undo | M-A Set Mark |
| ^X Exit | ^R Read File | ^_ Replace  | ^U Paste | ^J Justify | ^_ Go To Line | M-E Redo | M-6 Copy     |


```

```
GNU nano 7.2                                prog006.sh
ls -d */
;;
3)
    echo "Enter directory name to delete:"
    read dirname
    rmdir "$dirname"
    if [ $? -eq 0 ]; then
        echo "Directory '$dirname' deleted successfully."
    else
        echo "Error deleting directory (may not be empty)."

|         |              |             |          |            |               |          |              |
|---------|--------------|-------------|----------|------------|---------------|----------|--------------|
| ^G Help | ^O Write Out | ^W Where Is | ^K Cut   | ^T Execute | ^C Location   | M-U Undo | M-A Set Mark |
| ^X Exit | ^R Read File | ^_ Replace  | ^U Paste | ^J Justify | ^_ Go To Line | M-E Redo | M-6 Copy     |


```

```

root@DESKTOP-UU3FNJ4: ~
root@DESKTOP-UU3FNJ4: ~# nano prog007.sh
root@DESKTOP-UU3FNJ4:~# chmod +x prog007.sh
root@DESKTOP-UU3FNJ4:~# ./prog007.sh
Lab 7: Memory Usage Monitor

Current Memory status (using free command):
      total        used        free      shared  buff/cache   available
Mem:    7.6Gi       589Mi       7.0Gi       4.8Mi       186Mi       7.0Gi
Swap:    2.0Gi          0B       2.0Gi

Detailed Memory Info (using top command):

Available Memory:
./prog007.sh: line 15: Free: command not found

```

```

GNU nano 7.2                                prog007.sh
#!/bin/bash

echo "Lab 7: Memory Usage Monitor"

echo ""
echo "Current Memory status (using free command):"
free -h

echo ""
echo "Detailed Memory Info (using top command):"
top -b -n 1 | grep "Mib Mem"

echo ""
echo "Available Memory:"
Free -h | grep "Mem" | awk '{print "Available: " $7}'

[ Read 16 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line  M-E Redo      M-6 Copy

```

```
root@DESKTOP-UU3FNJ4: ~  
root@DESKTOP-UU3FNJ4:~# nano prog008.sh  
root@DESKTOP-UU3FNJ4:~# chmod +x prog008.sh  
root@DESKTOP-UU3FNJ4:~# ./prog008.sh  
Lab 8: Environment Variables Demo  
  
Local Variable:  
Name: Akshari  
Course: Operating System  
  
Using printenv command:  
Akshari  
Operating System  
  
All Environment Variables (using env):  
SHELL=/bin/bash  
WSL2_GUI_APPS_ENABLED=1  
WSL_DISTRO_NAME=Ubuntu  
WT_SESSION=18766da8-58cb-4d81-97c0-41c7843594e9  
NAME=DESKTOP-UU3FNJ4  
PWD=/root  
LOGNAME=root  
HOME=/root  
LANG=C.UTF-8  
WSL_INTEROP=/run/WSL/635_interop  
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*tar=01;31:*tgz=01;31:*arc=01;31:*arj=01;31:*taz=01;31:*lha=01;31:*lz4=01;31:*lzh=01;31:*lzma=01;31:*tlz=01;31:*txz=01;31:*tzo=01;31:*t7z=01;31:*zip=01;31:*z=01;31:*dz=01;31:*gz=01;31:*lrz=01;31:*lz=01;31:*lzo=01;31:*xz=01;31:*zst=01;31:*tzt=01;31:*bz2=01;31:*bz=01;31:*tbz=01;31:*tbz2=01;31:*tz=01;31:*deb=01;31:*rpm=01;31:*jar=01;31:*war=01;31:*ear=01;31:*sar=01;31:*rar=01;31:*alz=01;31:*ace=01;31:*zoo=01;31:*cpio=01;31:*7z=01;31:*rz=01;31:*cab=01;31:*wim=01;31:*swm=01;31:*dwm=01;31:*esd=01;31:*avif=01;35:*jpg=01;35:*jpeg=01;35:*mjpg=01;35:*mjpeg=01;35:*gif=01;35:*bmp=01;35:*pbm=01;35:*pgm=01;35:*ppm=01;35:*tga=01;35:*xbm=01;35:*xpm01;35:*m2v=01;35:*mkv=01;35:*webm=01;35:*webp=01;35:*ogm=01;35:*mp4=01;35:*m4v=01;35:*mp4v=01;35:*vob=01;35:*qt=01;35:*nuv=01;35:*wmv=01;35:*asf=01;35:*rm=01;35:*rmvb=01;35:*flc=01;35:*avi=01;35:*fli=01;35:*flv=01;35:*gl=01;35:*dl=01;35:*xcf=01;35:*xwd=01;35:*yuv=01;35:*cgm=01;35:*emf=01;35:*ogv=01;35:*ogx=01;35:*aac=00;36:*au=00;36:*flac=00;36:*m4a=00;36:*mid=00;36:*midi=00;36:*mka=00;36:*mp3=00;36:*mpc=00;36:*ogg=00;36:*ra=00;36:*wav=00;36:*oga=00;36:*opus=00;36:*spx=00;36:*xspf=00;36:*~=00;90:*#=00;90:*bak=00;90:*crdownload=00;90:*dpkg-dist=00;90:*dpkg-new=00;90:*dpkg-old=00;90:*dpkg-tmp=00;90:*old=00;90:*orig=00;90:*part=00;90:*rej=00;90:*rpmnew=00;90:*rpmorig=00;90:*rpmsave=00;90:*swp=00;90:*tmp=00;90:*ucf-dist=00;90:*ucf-new=00;90:*ucf-old=00;90:  
WAYLAND_DISPLAY=wayland-0  
LESSCLOSE=/usr/bin/lesspipe %s %s  
TERM=xterm-256color  
LESSOPEN=| /usr/bin/lesspipe %s  
USER=root  
DISPLAY=:0  
SHLVL=2  
XDG_RUNTIME_DIR=/run/user/0/  
WSLENV=WT_SESSION:WT_PROFILE_ID:  
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop  
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/Sougandhika S L/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/Sougandhika S L/AppData/Local/Programs/Python/Python314:/mnt/c/Users/Sougandhika S L/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Sougandhika S L/AppData/Local/Programs/Microsoft VS Code/bin:/snap/bin  
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/0/bus  
HOSTTYPE=x86_64  
course=Operating System  
PULSE_SERVER=unix:/mnt/wslg/PulseServer  
WT_PROFILE_ID={51855cb2-8cce-5362-8f54-464b92b32386}  
name=Akshari  
=/usr/bin/env
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
GNU nano 7.2 prog008.sh
#!/bin/bash

echo "Lab 8: Environment Variables Demo"

# setting variable
name="Akshari"
course="Operating System"

echo ""
echo "Local Variable:"
echo "Name: $name"
echo "Course: $course"

# Exporting variables
export name
export course

echo ""
echo "Using printenv command:"
printenv name
printenv course

echo ""
echo "All Environment Variables (using env):"
env

[ Read 25 lines ]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^_ Replace ^U Paste ^J Justify ^_/ Go To Line M-E Redo M-6 Copy
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
root@DESKTOP-UU3FNJ4:~# nano prog009.sh
root@DESKTOP-UU3FNJ4:~# chmod +x prog009.sh
root@DESKTOP-UU3FNJ4:~# ./prog009.sh
Lab 9: I/O Redirection Demo
Content written to file1.txt
Content appended to file1.txt

Reading from file1.txt using input redirection:
Hello, this is OS Lab
Appending new line

Trying to read non-existing file:
Error message stored in error.txt

Error file content:
cat: file_not_exist.txt: No such file or directory
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
GNU nano 7.2 prog009.sh
#!/bin/bash

echo "Lap 9: I/O Redirection Demo"

# Output redirection (overwrite)
echo "Hello, this is OS Lab" > file1.txt
echo "Content written to file1.txt"

# Append output
echo "Appending new line" >> file1.txt
echo "Content appended to file1.txt"

# Input redirection
echo ""
echo "Reading from file1.txt using input redirection:"
cat < file1.txt

# Error redirection
echo ""
echo "Trying to read non-existing file:"
cat file_not_exist.txt 2> error.txt
echo "Error message stored in error.txt"

# Display error file
echo ""
echo "Error file content:"
cat error.txt

[ Read 27 lines ]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-6 Copy
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v
root@DESKTOP-UU3FNJ4:~# nano prog0010.sh
root@DESKTOP-UU3FNJ4:~# chmod +x prog0010.sh
root@DESKTOP-UU3FNJ4:~# ./prog0010.sh
Command Piping Demo
Sample file 'fruits.txt' created with content:
apple
banana
apple
orange
banana
apple

Total lines in fruits.txt:
Number of times 'apple' appears: 3
Sorted unique fruits:
apple
banana
orange
```

```
root@DESKTOP-UU3FNJ4: ~ X root@DESKTOP-UU3FNJ4: ~ X + v prog0010.sh
GNU nano 7.2
#!/bin/bash
# Filename: pipe_demo.sh
# purpose: Demonstrate commad piping in Unix shell

echo "Command Piping Demo"

# 1. Create a sample file
echo -e "apple\nbanana\napple\norange\nbanana\napple" > fruits.txt
echo "Sample file 'fruits.txt' created with content:"
cat fruits.txt
echo

# 2. Count total lines using piping
total_lines=$(cat fruits.txt | wc -l)
echo "Total lines in fruits.txt: $total_lines"

# 3. Count occurrences of 'apple'
apple_count=$(cat fruits.txt | grep "apple" | wc -l)
echo "Number of times 'apple' appears: $apple_count"

# 4. Sort the file and remove duplicates
echo "Sorted unique fruits:"
cat fruits.txt | sort | uniq
echo

[ Read 24 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-6 Copy
```